Annex table 1 Improvements in nutritional status and recovery and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food type	Improvement in nutritional status or recovery	Factors that affect nutritional recovery
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUTF	328 (62.2%) recovered. **Measurement**: BMI>=18.5 kg/m^2 in two subsequent measurements	The factors that were associated with nutritional recovery were: - female (adjusted odds ratio (AOR) and 95% confidence interval (CI)= 2.16 (1.340-3.473)) - educated (AOR and 95%= 1.82(1.110-2.983)) - moderate undernutrition (AOR and 95%= 2.594 (1.251-5.377)) - mild undernutrition (AOR and 95% =14.25 (7.074-28.742)) - WHO stage I and II (AOR and 95% CI =3.345 (1.652-6.772)) - no opportunistic infection during admission (AOR and 95% =2.791(1.047-7.439))
Mallewa, et al 2018 31, Africa	Randomized clinical trial	-Adults -At least five years with HIV infection and CD4 cell count fewer than -100	1805 -897 to treatment -908 to non RUSF	Peri-urban regional hospital **Food type**: RUTF and RUSF	Adults and adolescents aged 13 years and older in the RUSF group had significantly greater gains in BMI, and MUAC than the no-RUSF group	Not assessed

		-Starting ART at the health facilities			(p=0·004, and 0·03, respectively). **Measurement:** Improvement in weight, BMI and MUAC (no cut off point)	
Nagata et al 2014 40, Kenya	Retrospective cohort study	-non pregnant adult patients -BMI less than 18.5 kg/m^2	1017	Rural village **Food type**: fortified corn-soy flour supplements	On average, enrolled clients gained 2.01 kg in weight and 0.73 kg/m^2 in BMI over follow-up (mean 100 days) with the highest gain among those severely undernourished. - Only 13.1% nutritional recovery according to criteria **Measurement**: BMI.20 kg/m^2	The factors associated with BMI >20 Kg/m^2 were - Greater BMI at baseline 1.52 (1.23–1.86) - younger age, 0.97 (0.94–1.00) - male gender, (0.51 (0.30–0.86)) - not requiring highly active antiretroviral therapy (HAART) (4.18 (2.42–7.22)

Ndekha et al 2009 34, Malawi	Randomised control trial	- Adults living with HIV - BMI<18.5 Kg/m^2	491	Clinic based **Food type**: fortified corn-soy flour supplements	After 14 weeks, patients receiving fortified spread had a greater increase in BMI and fat-free body mass than those receiving corn-soy blend: 2.2 (SD 1.9) v 1.7 (SD 1.6). Measurement: Gain in BMI (no cut off point)	No significant differences in the CD4 count, HIV viral load, assessment of quality of life, or adherence to antiretroviral therapy were noted between the two groups
Oguntibeju et al 2007 41, South Africa	Retrospective cohort study	- non pregnant adults with HIV, non-diabetic and non-hypertensive - not on ART - CD4 cell count 100-350 cells/mm^3	35	Not specified **Food type**: extract of hypoxis, grape fruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green	19 (67.9%) of the 28 patients had a BMI within the normal range after nutrient intervention **Measurement:** BMI>=18.5kg/m^2	- significant positive correlation between the BMI and fat percentage (statistics not reported) - CD4+T cell count showed no correlation with any of the anthropometric indices - high viral load has significant negative correlation with the lean body mass and BMI (statistics not reported)

				leaf extract, folic acid and natural antioxidant (biocydin)		
Olsen et al 2014 33, Ethiopia	Randomised control trial	- Age greater than 18 years - BMI greater than 16 Kg/m^2	453	Referral hospital **Food type**: Plump sup	At three months, participants receiving the supplements containing whey or soy had increased their lean body mass by 0.85 kg (95% confidence interval 0.16 kg to 1.53 kg) and 0.97 kg (0.29 kg to 1.64 kg), respectively, more than controls. **Measurement**: Weight gain and BMI (no cut off point)	- No effect on physical activity - The effects of the two supplements, however, were not significantly different in direct comparison
Rawat et al, 2014 35, Uganda	Prospective cohort	- Food insecure people living with HIV who are new to ART. - employment status and income	904	Community based **Food type**: WFP Rations	Over 12 months, food assistance significantly increased body mass index by 0.6 kg/m2 (P, 0.01) and mid-upper arm	- no impact on CD4 count, Hb concentrations, or Individual Dietary Diversity Score - at the household level, food assistance increased the Household Food Insecurity Access Scale, by 2.1 points (P, 0.01) - food assistance increased BMI by 0.6 kg/m2 and MUAC by 6.7 mm.

		- housing characteristics and past experience of food insecurity			circumference by 6.7 mm (P, 0.05). **Measurement:** Change in BMI (no cut off points)	- no statistically significant impact of food assistance on Hb concentration or CD4 count
Sadler, et al, 25 Ethiopia	Comparative follow up study. Report, not peer reviewed	Adults living with HIV - BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres **Food type**: RUTF/RUSF	636 (32.6) graduated in the intervention group and 120 (18.8) in the comparison group **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Factors that were associated with nutritional recovery were: - being in the intervention group 2.4 (1.6-3.4) - female 1.4 (1.0-1.9) - attending in a hospital 0.5 (0.4-0.7)
Scarcella. et al 43, Mozambique	Randomised control trial	- Adult HIV patients - BMI <18.5 Kg/m^2 - presence of wasting - TB co-infection - insufficient access to food - CD4 count < 200	106	Public Health Centre **Food type**: - food package consisted of a ration of cereals (rice or corn) of around 15kg; pulses (5kg); peanuts (3kg); sugar (2kg); sunflower seed oil (3Lt.);	In the cohort not in program the mean BMI values showed an increase but not significant (initial value: 21.9±2.9; final value: 22.5±3.3) **Measurement**: Increase in BMI (no cut off point)	- haemoglobin showed an increase but not significant - in the intervention group both, BMI and Haemoglobin shown a significant increase - increase in BMI was statistically associated with the dietary diversity score in the intervention group

				at least 7kg of fortified corn soy blend		
Tesfay et al, 42 Ethiopia	Retrospective cohort study	- Adults living with HIV - BMI<18.5	1757	Hospitals **Food type**: RUTF/RUSF	55.3% (95% CI = 53.2–57.4) recovered from undernutrition **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Not assessed
van Oosterhout et al 2010 37, Malawi	Retrospective cohort study	-Adults aged 18 years and older -BMI 18.5 kg/m^2 -ART naïve or - WHO clinical stage III or IV Or - CD4 count B250 cells/ml irrespective of clinical stage	593	Centeral hospital **Food type**: corn/soy blended flour (CSB) or ready-to-use fortified spread (RUFS), a peanut butter-based product	- Linear regression analysis showed that increase in BMI was greatest in the RUSF group and better in the CSB group than in those receiving no food supplementation at 14 weeks. - These differences were no longer significant at 26 weeks **Measurement:** Increase in BMI	- no longer a significant effect of supplementary feeding on an increase in BMI, 12 weeks after the feeding was stopped. - factors that were associated with nutritional recovery or improved BMI was not assessed.

Annex table 2: Shows program failure (default + nonresponse +death) after enrolment in a nutritional program in sub Saharan Africa

Author, year and country	Study design	Study population	N	Setting and type of food	Nutritional program failure	Factors that affect nutritional program failure
Ahou et al 2011 38, Kenya and uganda	Retrospective cohort study	Adults aged 15 years and older living with HIV BMI less than 17 kg/m^2 and bilateral pitting oedema	1106	Regional hospital Type of food: RUTF/RUSF	An overall total of 531 of (48.0%) patients failed nutrition therapy	Factors that were associated with program failure were: - male gender (OR = 1.5, 95% CI 1.2-2.0), - patients with severe undernutrition at nutrition programme enrolment (OR = 2.2, 95% CI 1.7-2.8), and - those never started on ART (OR = 4.5, 95% CI 2.7-7.7) for those eligible; OR = 1.6, 95% CI 1.0-2.5 for those ineligible for ART at enrolment
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital Type of food: RUTF/RUSF	Overall program failure rate was 196 (37.6%)	The factors that were associated with nutritional program failure were not assessed.

Annex table 3: Nonresponse to the nutritional program and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food	Nonresponse to the nutritional program	Factors that contribute to non-response to the nutritional program
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUFT/RUSF	- 155 (29.6%) non-respondent - **Measurement**: not reaching 18.5 kg/m^2	The factors that affect non-response were not assessed
Sadler, et al , 25 Ethiopia	Comparative follow up study. not peer reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each. 1956 and 639 were including in the final analysis	Health facilities / health centres **Food type**: RUFT/RUSF	- 1316 (67.4%) non-respondent in the intervention group and 519 (81.2) become non-respondent in the comparison group - **Measurement**: not reaching 18.5 kg/m^2	Factors that were associated with nutritional non-response were: - attending the program at a hospital was associated with an increased likelihood of non-response (OR=1.75, p=0.021), - severe food insecurity was associated with a decreased likelihood of non-response (OR=0.46, p=0.016) compared to those who were “food secure.
Tesfay et al , 42 Ethiopia	Retrospective cohort study	Adults living with HIV and-BMI<18.5	1757	Hospitals **Food type**: RUFT/RUSF	- 21% (95% CI = 19.7–23.4) completed the program but failed to recover from undernutrition. - **Measurement**: not reaching 18.5 kg/m^2	- urban residence, (AOR = 1.46, 95% CI = 1.4–2.91), - attending Shul (AOR = 2.92, 95% CI = 2.04–4.19) and - Lemlem Karl (AOR = 1.49, 95% CI 1.05–2.11) hospitals

Annex table 4: Default from the nutritional program and its determinants in sub-Saharan Africa

Author, year, and country	Study design	Study population	N	Setting	Default to the nutritional program	Factors that contribute to default to the nutritional program
Sadler, 25 et al , Ethiopia	Comparative follows up study. not peer-reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres	70.6% defaulted from the nutritional program among the intervention group	- severe acute undernutrition (OR: 1.89, P-value=0.003)
Tesfay et al 42 , Ethiopia	Retrospective cohort study	Adults living with HIV and BMI<18.5	1757	Hospitals	19% (95% CI, 17.3–20.7) did not complete the program.	Factors associated with program incompleteness. - living in urban areas (AOR = 1.44, 95% CI = 1.05– 1.97), - employed (AOR = 1.39, 95% CI = 1.01–1.93), - attending Shul (AOR = 4.6, 95% CI = 3.15–6.71) and Lemlem Karl (AOR = 2.5, 95% CI = 1.69–3.71) hospitals, - in clinical stages II (AOR = 2.49, 95% CI = 1.59–3.91) and III (AOR = 1.46(1.02–2.07), - on ART for less than six months (AOR = 1.61, 95% CI = 1.09–2.39), - anaemic (AOR = 1.77, 95% = 1.29–2.41), and - severe acute undernutrition at enrolment (AOR = 6.43, 95% CI = 4.69–8.3)